Famicom JAMMA board

power

File: power.kicad_sch

Power: +5V entry, switched +5V_SW, +3V3, supply of the RP2A03 modules

cpu

File: cpu.kicad_sch

CPU: RP2A03, work RAM, address decoder, data-bus resistors

ppu

File: ppu.kicad_sch

PPU: RP2C02, address latch, video RAM, sprite detection

cart

File: cart.kicad_sch

Cartridge connector and the dump circuit for emulation mode

video

File: video.kicad_sch

Video: RGB resistor DAC and composite sync

audio

File: audio.kicad_sch

Audio: mixer, loop through the cartridge, power amplifier

jamma

File: jamma.kicad_sch

JAMMA edge connector, inputs, coin counter and lockout drivers

mcu

File: mcu.kicad_sch

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

PWR_FLAG

GND

PWR_FLAG

+5V

PWR_FLAG

+5V_SW

PWR_FLAG

VSYS

Sheet: /
File: fcjamma.kicad_sch

Title: Famicom JAMMA board

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/9

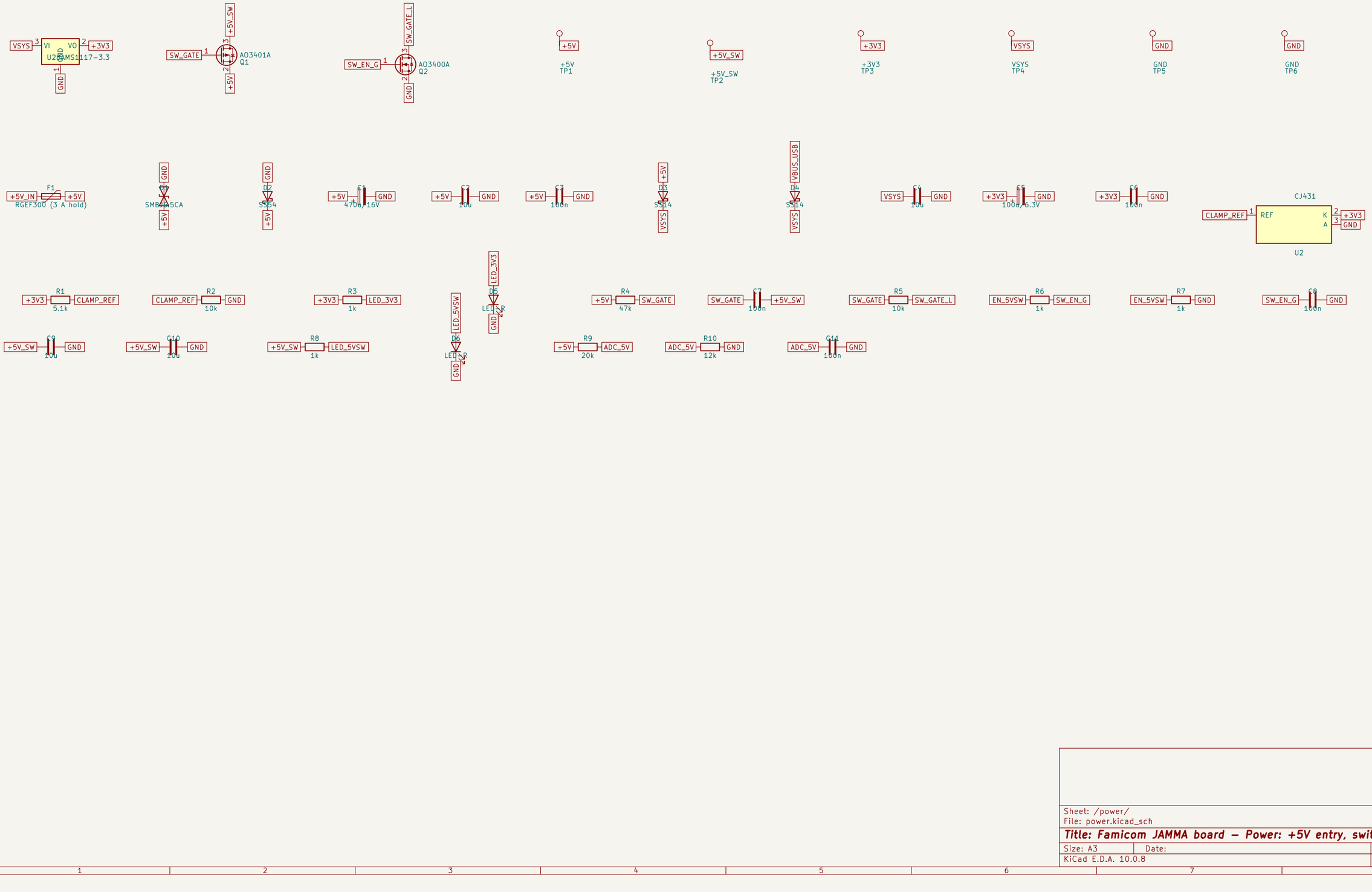
Power: +5V entry, switched +5V_SW, +3V3, supply of the Pico modules

+5V_IN comes from the JAMMA fingers 3, 4, C, D. −5 V and +12 V are not used.

+5V_SW feeds the CPU, the PPU, the memories, the 5 V logic and the cartridge. It is off until Pico C raises EN_5VSW,

so that no 5 V signal reaches a Pico pin before that Pico is powered (the RP2350 pins tolerate 5 V only while powered).

VSYS feeds the three Pico modules and the +3V3 regulator from +5V or from the USB socket, whichever is present.



Sheet: /power/
File: power.kicad_sch

Title: Famicom JAMMA board — Power: +5V entry, switched +5V_SW, +3V3, s

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 2/9

CPU: RP2A03, work RAM, address decoder, data-bus resistors

U1 is the donor's RP2A03 in a 40-pin socket. In emulation mode the socket is empty and the dump shift registers drive the buses.

The 470 ohm resistors let Pico B replace what the PPU sees on a write without disturbing the CPU side of the bus.

No pull-ups on the CPU data bus: they are known to upset some PPU revisions, and games read the floating bus.

RP2A03 (donor)

M68AF127B (work RAM)

74HC13B

74HC00

CPU bus (option)

RAM A11–A16 (option)

Sheet: /cpu/		
File: cpu.kicad_sch		
Title: Famicom JAMMA board — CPU: RP2A03, work RAM, address decoder, da		
Size: A2	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 3/9

PPU: RP2C02, address latch, video RAM, sprite detector

U6 is the donor's RP2C02 in a 40-pin socket. EXT0–3 give the lower four bits of the palette index of every pixel once Pico B has forced bit 6 of register 2000 high. The fifth bit (sprite or background) is read from the brightness of VOUT: Pico B makes the PPU store a dark colour in every background entry and a bright one in every sprite entry.

RP2C02 (donor)

U6

74HC373

U7

M68AF127B (video RAM)

U8

TL712CP

U9

PPU bus (option)

J4

CMP_OUT

CMP_OUT
TP11

VBUF

VBUF
TP9

VREF

VREF
TP10

MMBT3904
Q3

PPU /RST = +5V (Famicom PPU)

JP1

composite video (test)

J5

sprite bit source

J6

Sheet: /ppu/	
File: ppu.kicad_sch	
Title: Famicom JAMMA board – PPU: RP2C02, address latch, video RAM, sprit	
Size: A2	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 4/9

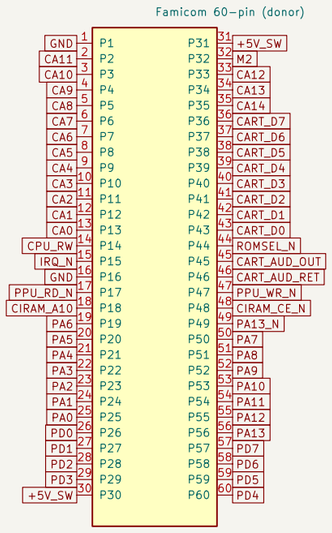
Cartridge connector and the dump circuit for emulation mode

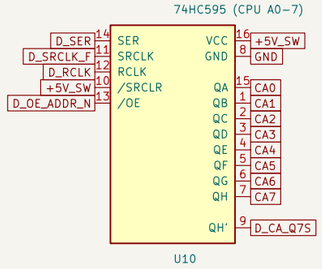
J8 is the 60-pin connector taken from the donor Famicom. Pin 45 carries the board's sound to the cartridge, pin 46 brings it back.

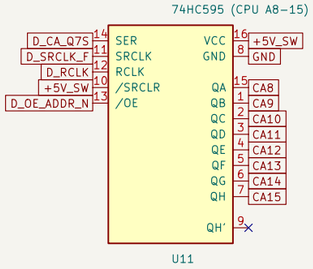
Dump circuit: with the CPU and PPU sockets empty, Pico C reads the cartridge through shift registers and hands the data to Pico A.

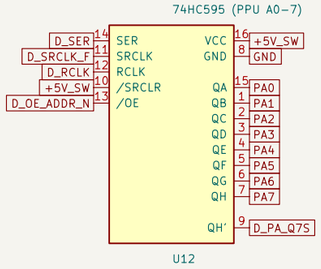
The six 74HC595 run from +5V_SW; Pico C drives their inputs open-drain against 4.7 k pull-ups, so they see proper 5 V levels.

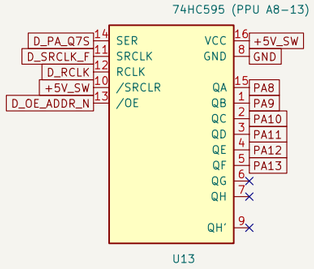
In real-chip mode D_OE_ADDR_N and D_OE_DATA_N stay high (pull-ups) and every 74HC595 output is switched off.

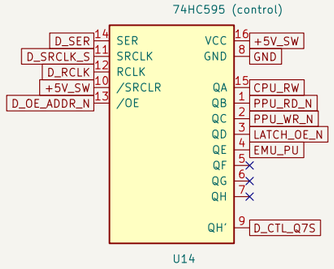


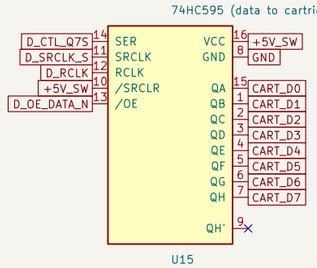


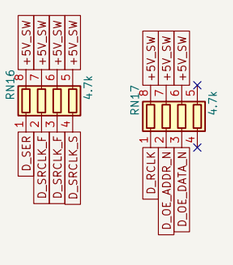


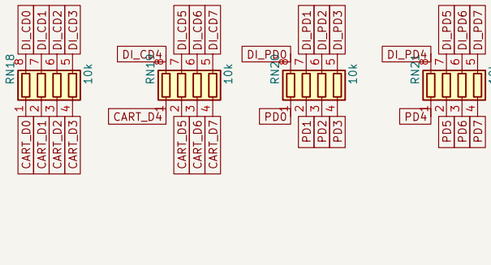


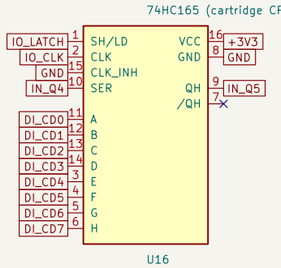


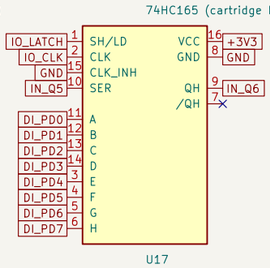
















Sheet: /cart/
File: cart.kicad_sch

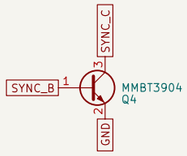
Title: Famicom JAMMA board – Cartridge connector and the dump circuit for

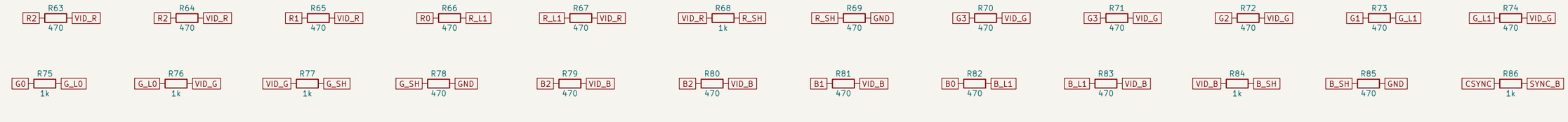
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 5/9

Video: RGB resistor DAC and composite sync

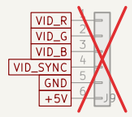
Pico A drives the DAC directly. Full scale is 3.0 V open circuit, about 2.85 V into 2 k, source resistance about 125 ohm.

Red and blue have 3 bits, green has 4. Only 470 ohm and 1 k are used so that the parts list stays short.





RGB + sync (monitor / option)



Sheet: /video/
File: video.kicad_sch

Title: Famicom JAMMA board – Video: RGB resistor DAC and composite sync

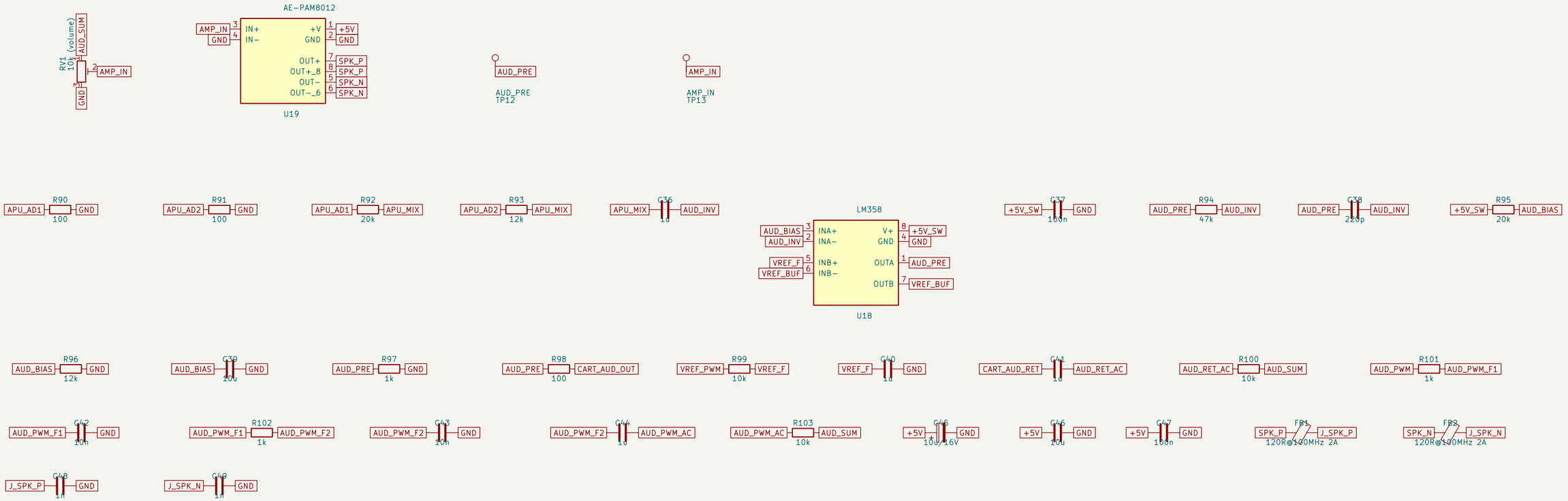
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 6/9

Audio: mixer, loop through the cartridge, power amplifier

Real-chip mode: 2A03 pins -> mixer (Famicom values) -> cartridge pin 45 -> cartridge -> pin 46 -> volume -> amplifier.

Emulation mode: Pico A's PWM output -> two RC stages -> volume -> amplifier.

The loudspeaker outputs are bridge-tied: JAMMA L is not ground.



Sheet: /audio/		
File: audio.kicad_sch		
Title: Famicom JAMMA board – Audio: mixer, loop through the cartridge, pow		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 7/9

JAMMA edge connector, inputs, coin counter and lock-out drivers

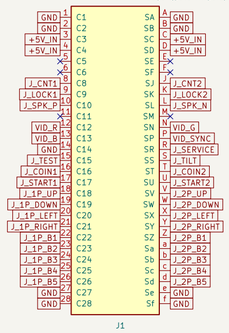
Every input: 1 k pull-up to +3V3 (3.3 mA through the switch), 4.7 k in series, 10 nF to ground, then a 74HC165 on +3V3.

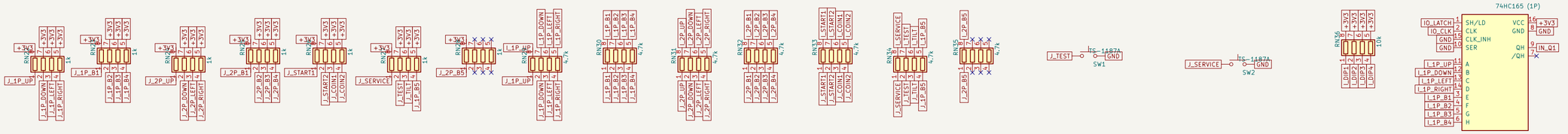
Pico C reads the chain: after a low pulse on IO_LATCH the first bit on IN_DATA is input H of U17, the last is input A of U20.

The same pulse copies the 74HC595's shift register to its outputs, so one cycle reads the inputs and writes the outputs.

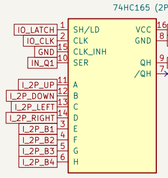
Fingers 5/E (−5 V), 6/F (+12 V) and 11/M are not connected. 7/H is the key slot.

JAMMA edge (gold fingers)

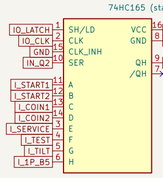




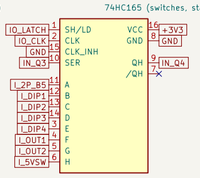
74HC165 (2P)



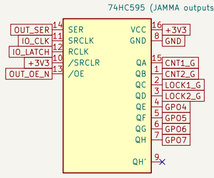
74HC165 (start, coin, service)

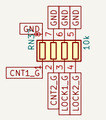


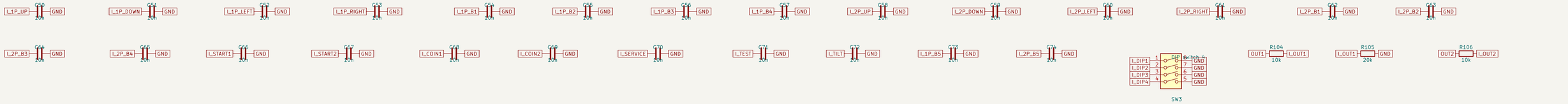
74HC165 (switches, status)

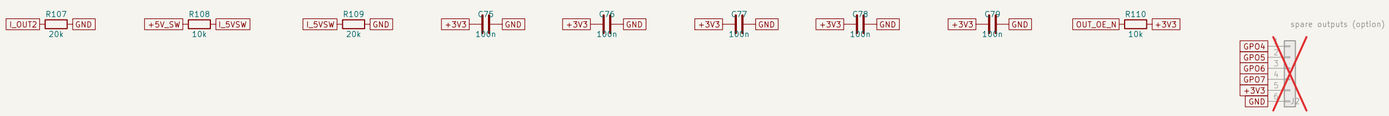


74HC595 (JAMMA outputs)









Sheet: /jamma/
File: jamma.kicad_sch

Title: Famicom JAMMA board — JAMMA edge connector, inputs, coin counter a

Size: A2	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 8/9

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

Pico A (video): clock and reset for CPU and PPU, palette index and sprite bit in, RGB and sync out; the emulator in emulation mode.

Pico B (bus): watches the CPU's accesses to the PPU, replaces palette data on the PPU side, answers the controller reads.

Pico C (input/output and management): JAMMA inputs and outputs, coin and timer logic, cartridge dump, USB Type-C,

power switch of the Famicom half, and reprogramming of Pico A and Pico B through their debug pins (J13, J14).

Pico 2 A (video)

A_EXT0 A_EXT1 A_EXT2 A_EXT3 A_SPR A_ALE R0 R1 R2 G0 G1 G2 G3 B0 B1 B2

GP0 GP1 GP2 GP3 GP4 GP5 GP6 GP7 GP8 GP9 GP10 GP11 GP12 GP13 GP14 GP15

VSYS VBUS 3V3 3V3_EN ADC_VREF RUN GP16 GP17 GP18 GP19 GP20 GP21 GP22 GP26 GP27 GP28 AGND GND_3 GND_8 GND_13 GND_18 GND_23 GND_28 GND_38

VSYS A_3V3 RUN_A CSYNC CLK A_PPU_RST A_MISO C_MOSI LINK_BA C_SCK A_CPU_RST AUD_PWM VREF_PWM GND

U25

A_3V3

A_3V3
TP14

Pico 2 B (bus)

B_CD0 B_CD1 B_CD2 B_CD3 B_CD4 B_CD5 B_CD6 B_CD7 B_OVR0 B_OVR1 B_OVR2 B_OVR3 B_OVR4 B_OVR5 B_OVR6 B_OVR7

GP0 GP1 GP2 GP3 GP4 GP5 GP6 GP7 GP8 GP9 GP10 GP11 GP12 GP13 GP14 GP15

VSYS VBUS 3V3 3V3_EN ADC_VREF RUN GP16 GP17 GP18 GP19 GP20 GP21 GP22 GP26 GP27 GP28 AGND GND_3 GND_8 GND_13 GND_18 GND_23 GND_28 GND_38

VSYS B_3V3 RUN_B B_CA0 B_CA1 B_CA2 B_RW B_PPUCS B_OE1 B_OE2 B_OUT0 LINK_CB B_TX GND

U26

B_3V3

B_3V3
TP15

Pico 2 C (I/O, management)

D_SER D_SRCLK_F D_SRCLK_S D_RCLK D_OE_ADDR_N D_OE_DATA_N C_M2 OUT_OE_N IN_DATA IO_LATCH IO_CLK OUT_SER C_SWCLK_A C_SWDIO_A C_SWCLK_B C_SWDIO_B

GP0 GP1 GP2 GP3 GP4 GP5 GP6 GP7 GP8 GP9 GP10 GP11 GP12 GP13 GP14 GP15

VSYS VBUS 3V3 3V3_EN ADC_VREF RUN GP16 GP17 GP18 GP19 GP20 GP21 GP22 GP26 GP27 GP28 AGND GND_3 GND_8 GND_13 GND_18 GND_23 GND_28 GND_38

VSYS C_3V3 RUN_C C_TX C_RUN_A C_SCK_O C_MOSI_O C_MISO C_RUN_B EN_5VSW ADC_5V C_CIRAM_A10 C_SPARE GND

U27

C_3V3

C_3V3
TP16

RN38 SWCLK_A SWDIO_A SWCLK_B SWDIO_B 1k

C_SWCLK_A C_SWDIO_A C_SWCLK_B C_SWDIO_B

RN39 RUN_A RUN_B 1k

C_RUN_A C_RUN_B

RUN_C TS-1187A GND SW4

C_SPARE

C_SPARE
TP17

C80 VSYS GND 10u

C81 VSYS GND 10u

C82 VSYS GND 10u

R111 B_TX LINK_BA 100

R112 C_TX LINK_CB 100

R113 C_SCK_O C_SCK 100

R114 C_MOSI_O C_MOSI 100

R115 A_MISO C_MISO 100

debug pins of Pico A

SWCLK_A GND SWDIO_A

J13

debug pins of Pico B

SWCLK_B GND SWDIO_B

J14

debug pins of Pico C

SWCLK_C GND SWDIO_C

J15

debug probe for Pico C (option)

SWCLK_C GND SWDIO_C

J16

AE-USB2.0-TYPE-C

VBUS_USB USB_DM USB_DP USB_CC1 USB_CC2

VBUS D- D+ CC1 CC2 GND GND_7 GND GND

J10

R116 USB_CC1 GND 5.1k

R117 USB_CC2 GND 5.1k

USB to Pico C

VBUS_USB USB_DM USB_DP GND

J11

Sheet: /mcu/
File: mcu.kicad_sch

Title: Famicom JAMMA board – Three Raspberry Pi Pico 2 modules, USB Type

Size: A3 | Date: | **Rev: 0.1**

KiCad E.D.A. 10.0.8 | Id: 9/9